

Station-keeping for high-altitude balloon with reinforcement learning

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Abstract

The operating altitudes of the stratospheric balloons can reach near-space altitudes of more than 20 km, where the appearance of the quasi-zero wind layer follows the seasonal rule. As unmanned aerial vehicles with application potential, the effective flight control of balloon position is crucial. This research develops a station-keeping control approach based on reinforcement learning, and the control strategy also considers the characteristics of the local wind field. Firstly, an atmospheric environment model with an uncertain wind field is established according to the analysis of the historical wind data. The model serves as a training environment for the balloon station-keeping strategy training. Secondly, the thermal model, dynamic model, and altitude control model are introduced, and an environment based on historical real wind data is developed. Thirdly, the dueling double Q-learning deep network with prioritized experience replay method is applied to the station keeping of high-altitude balloons. The Priority Experience Replay based on High-Value Samples (HVS-PER) is developed to improve the stability of strategy training. Finally, the performance of the optimal network is evaluated by the total reward, horizontal displacement, and effective working time under the uncertain wind environment. The strategy analysis also has reference to exploiting appropriate initial positions and capturing the opportunity of releasing a balloon. This work confirms that the control strategy is viable in complex and variable wind field environments and is capable of long-duration flights.

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Keywords: Dueling double deep Q learning network; Model of the uncertain wind field; Reinforcement learning; Stratospheric balloon; Station-keeping strategy

1. Introduction

In recent years, high-altitude balloons have played an increasingly important role in earth observation, environmental monitoring, and communication relays (Yajima et al., 2009; Doi et al., 2018). The balloons operate in an environment with limited vertical convection, steady ambient temperature, and low electromagnetic radiation (Tobin et al., 2017), and they can float for several months without refueling or relying on land facilities for energy (González-Bárcena et al., 2020). Although the balloon is inexpensive and durable, the wind could blow it away from the target area. Thus, precise management of balloon location is

required for stable communication between airborne equipment and base stations (van Wynsberghe and Turak, 2016). Previous researches have proved that balloons cannot floating within the specified location without control (Cathey et al., 2017; Du et al., 2019a). Therefore, it is necessary to design a controller and actuators to realize the balloon position control.

To control the horizontal displacement of the high-altitude balloon, Newman (Newman, 2005) designed a kind of ultra-long-duration balloon equipped with an extra system, which includes the renewable energy system, fixtures, and propellers. This research indicates the importance of modeling and analysis of the propulsion device. In addition to traditional propulsion equipment such as motors and propellers, the balloon also can modify its flight trajectory using some simple auxiliary structures.

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Ramesh (Ramesh et al., 2017) presents that the balloon-stratosail system and dual-balloon can be used to keep the main balloon within the target area. Yoder (Yoder et al., 2018) designed a trajectory control system by using a PI controller for the scientific balloon with tether and sail. Zhang (Zhang et al., 2020) proposed the tandem tethered balloon system, which can exploit the opposite directions of the winds at different altitudes to reduce the horizontal displacement, and Saito (Saito et al., 2010) used the super-pressure balloon and payload to extend the capability of the tandem balloon system. As a low-cost and long duration flying platform, the balloons are expected to be more affordable, compact and simple to produce. Öznur used the methods of increasing or decreasing helium and counterweight to change floating altitude of the balloon (Kayhan and Hastaoglu, 2014). NASA's balloon round earth flight test proves that the over pressured balloon has excellent durability, which ensures the safety of long-term flight of the balloon (Fairbrother, 2015; Cathey et al., 2015). Hall (Hall et al., 2019) designed the structure with air ballast, it helps to reduce the complexity of the whole system and increase the reliability. Based on the balloon with helium bag and air ballast, Du (Du et al., 2019b) designed a strategy that adjusts the air mass in the ballast to reach the expected altitude where the wind direction is suitable for station keeping. These researches showed that the balloon can realize horizontal displacement by adjusting the altitude, and the combination of valves and airbag helps to reduce the complexity of the system. With the rapid advancement of meteorological observation, the vertical wind direction variety could be recorded as wind layer information, and the wind with varying directions inside these multiple wind layers can be utilized to modify the balloon's horizontal position.

Reinforcement learning (RL) can solve the decision optimization problem of sequential actions without labeling data. Collecting balloon flight data in various places and seasons is inconvenient. RL can be used to handle problems requiring balloon station-keeping based on atmospheric environment simulation. Based on the memory generated during the sampling process, the agent can prepare a strategy by training an optimizer for the possible changes in the test period. And the uncertainty of the wind can be described as the state transition probability (Fathpour et al., 2014; Hamada et al., 2018). As artificial intelligence and computer performance improve, RL method is employed in a variety of engineering domains. (Yang et al., 2020; Wu et al., 2020). Q-learning is a widely used RL methodology that is based on a model-free approach and well suited for control problems (Kiumarsi et al., 2018). The controller trained by the RL method can guide the actions according to the environment states (Hwangbo et al., 2017). Project Loon has realized a 39-day controlled experiment of the balloon over the Pacific Ocean with the reinforcement learning method. During the trail, the control strategy can take maximum advantage of the wind at different altitudes to realize the position con-

trol (Bellemare et al., 2020). This study develops the training framework based on the Dueling Network theory. The dueling Q network consists of two parts: the state value function and the state-dependent action advantage function. The dueling network has better performance than the full connect network because the dueling network can avoid strategy instability caused by too small a difference between different strategies during the training process (Ziyu Wang, 2016). Prioritized Experience Replay (PER) is used to prioritize and replay samples for the network, and the priorities of the samples are updated when the new samples are stored in the replay buffer (Schaul et al., 2015). The PER method specifies the sampling sequence during the network training and filters the samples in the training set buffer. Previous research has proved that PER can reduce training time and contribute to network convergence, because some samples contain better decisions for the mission and these experiences can help the networks learn the strategy efficiently (Ramacic and Bonarini, 2020).

ERA5 provides hourly estimates of a large number of atmospheric variables, including the historical data of the wind field at 137 levels from the surface to a height of 80 km (Hersbach et al., 2018). In the stratosphere, the wind layer that has a reversed zonal component and a low velocity in meridional is referred to as the quasi-zero wind layer. According to the ERA-40 reanalyzed wind dataset, the quasi-zero wind layer usually exists throughout the late spring and early summer at 5°N–32.5°N (Xiao, 2008). The wind velocity is normally less than 10 m/s when the quasi-zero wind layer exists, there are nevertheless variations in the short term in direction and speed (Hersbach et al., 2018). Wind-heading predictions or historical data errors greater than 90° are frequent, and are greatest at the equator (Friedrich et al., 2017). On the basis of historical data on local wind, a dynamic wind model can be developed. The characteristics of the local wind field changes can be summarized. The regularity is helpful to fit the discrete data through mathematical methods, and the environment model state does not need to contain the original discrete data. Present researches have provided some flexible model for the studies of wind direction, and the researches of Han and Masseran indicate the Von Mises distribution gives a better fit for the wind direction in reality (Han et al., 2018; Masseran et al., 2013). Fig. 1 presents the June wind field data in Changsha (28°N, 114°E) from 2019 to 2021. The average wind speeds of 20 km low, and the quasi-zero wind layers remain above Changsha for several days. During the date indicated in Fig. 1(b), it is suitable for the balloon station-keeping experiments.

In this paper, the historical wind field data proved that the uncertain wind field model can be described by the Von Mises distribution and bandwidth parameter in Changsha during June. Combined with the wind field analysis, we build an atmospheric environment model with an uncertain wind field for strategy training, and the probabil-

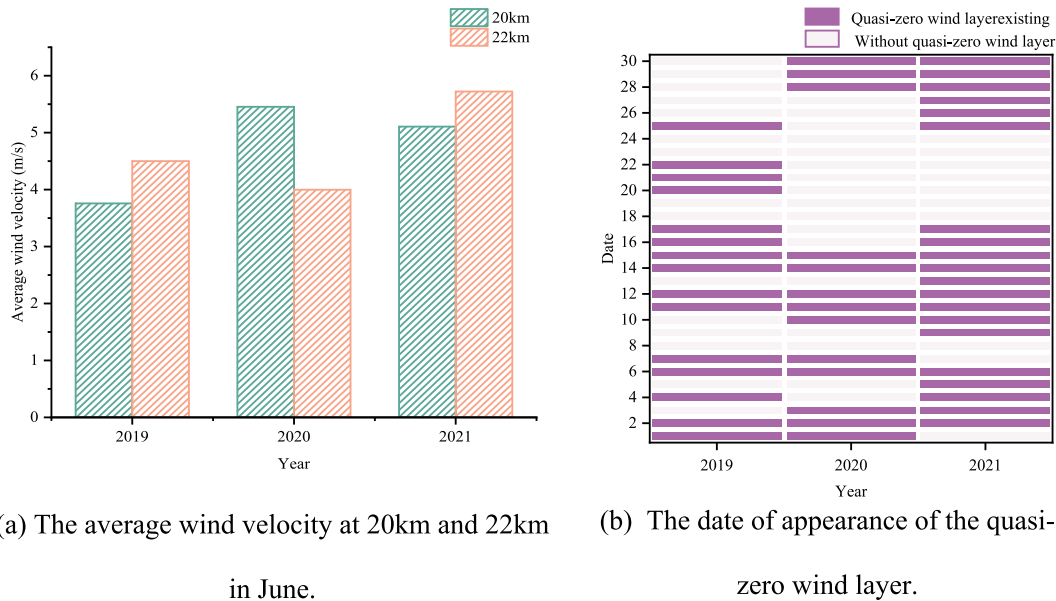


Fig. 1. Wind field statistics in Changsha (28°N, 114°E) from 2019 to 2021.

ity density function and bandwidth parameter are used to simulate the possible disturbances during training. In order to enhance the strategy stability and the utilization of high-value samples in the training process, Prioritized Experience Replay based on the High-Value Samples method (HVS-PER) is developed to reduce the time spent traversing the memory buffer. As for the deep neural network, the training framework takes the Double Dueling Deep Q-learning (D3QN) as the strategy optimizer, and the flow-chart of the framework is shown in Fig. 2.

Finally, the total reward, horizontal displacement, and effective working time are defined as the indexes to evaluate the performance of the controller. The test environment is created using historical and real-time wind field data, and the performance of the control strategy is evaluated over a range of bandwidth parameters. Moreover, this work also tests the influence of the initial position on the flight radius, which can become the reference for balloon launch.

2. Balloon theoretical model

2.1. Altitude control system in real wind field

The super-pressure balloon is a type of aircraft that can remain in the atmosphere for an extended period of time. The local atmosphere and the density of the balloon can have an effect on the altitude of the balloon. The horizontal wind can cause the balloon to move in the horizontal direction. Therefore, the balloon can shuttle through different wind layers in the vertical direction to control the horizontal flight direction. The quasi-zero wind layers typically emerge between 18 and 22 km in altitude and vary with the seasons (Tao et al., 2012).

To take advantage of the wind at different altitudes, the Altitude Control System (ACS) is designed to adjust its

altitude by pumping or venting air. The schematic diagram of the super-pressure balloon is represented in Fig. 3. The interior of the balloon is separated into two closed airbags by a membrane, and there are two valves for charging and discharging air. The pressure sensor can protect the balloon from bursting. When the valve discharges air, the pressure inside the balloon decreases which causes the helium bag to expand. Because the pressure within the super-pressure balloon is always greater than atmospheric pressure, the balloon's total volume and buoyancy remain constant. Therefore, combining the overall design and payload of the balloon, the gravity of the balloon can be adjusted by inflation or deflation of air bag inside the envelope and vertical motion can be realized. And the ACS can be used to select the ideal wind direction to control the horizontal trajectory of the balloon. The principle of the station-keeping process is shown in Fig. 4 with the ACS. The balloon adjusts its flight altitude around the quasi-zero layer to achieve horizontal reciprocating motion.

In addition, the thermal radiation from atmosphere, ground, sun and other sources can affect the temperature changes and overpressure of the balloon, which are the main restrictions of the adjustment range of the balloon altitude. To study the station-keeping performance of the balloon, four fundamental assumptions made in this paper are as follows (Zhu et al., 2019).

- (1) The atmosphere temperature, pressure and density are functions of geopotential altitude according to the US Standard Atmosphere Model (U.S. Standard Atmosphere, 1976) (Li et al., 2016).
- (2) The turbulent wind in stratosphere is ignored. The wind fields are steady for several days (Chiba et al., 2017).

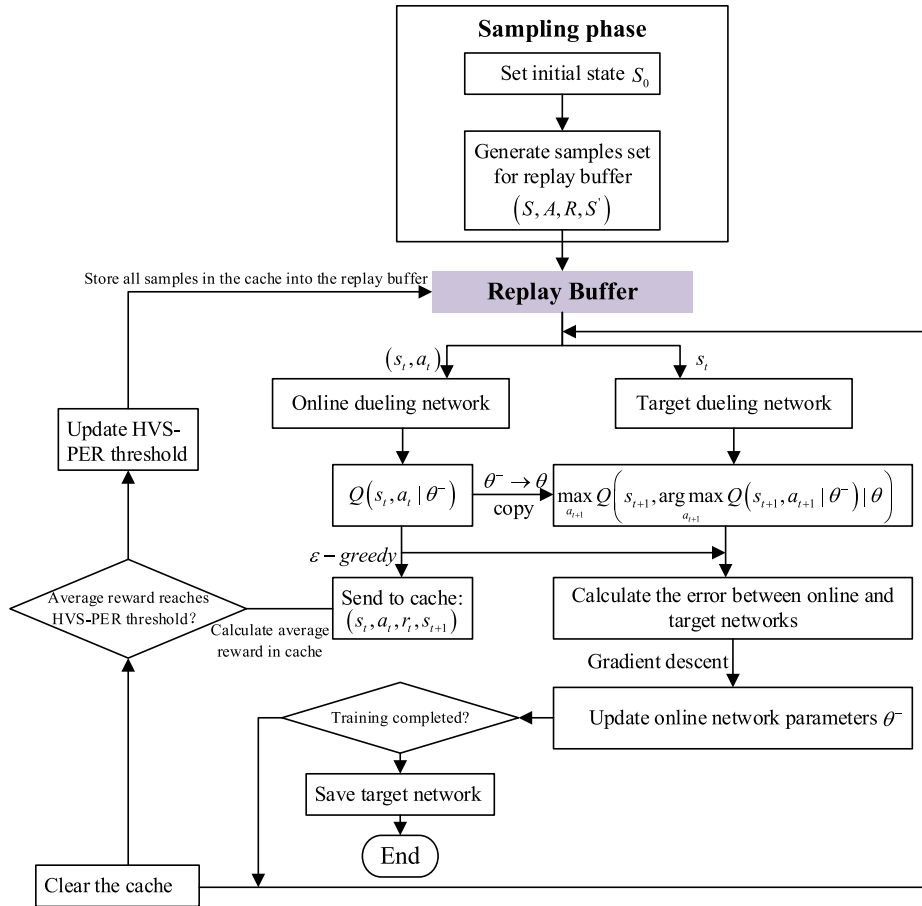


Fig. 2. Flowchart of the D3QN framework with HVS-PER method.

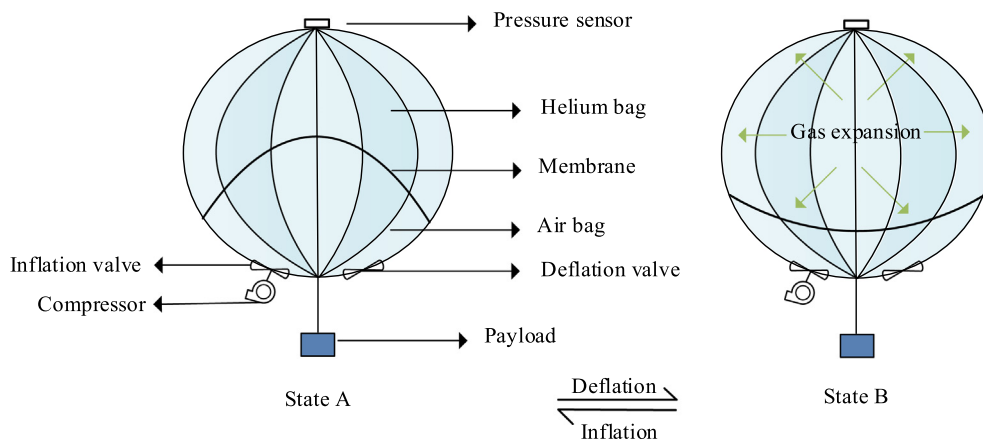


Fig. 3. Schematic diagram of super-pressure balloon with Altitude Control System.

- (3) The values of density, temperature and pressure of the Helium and the air are considered uniform inside the balloon.
- (4) The pressure inside the Helium capsule keeps same as the pressure inside the air capsule during both inflation and deflation. And all gas is assumed to follow the idea gas law.

2.2. Thermal model

The balloon movement is influenced by thermal environment (Aubin et al., 2017). Different source of thermal radiations from the atmospheric environment result in large diurnal variations in the temperature of the gas in the bal-

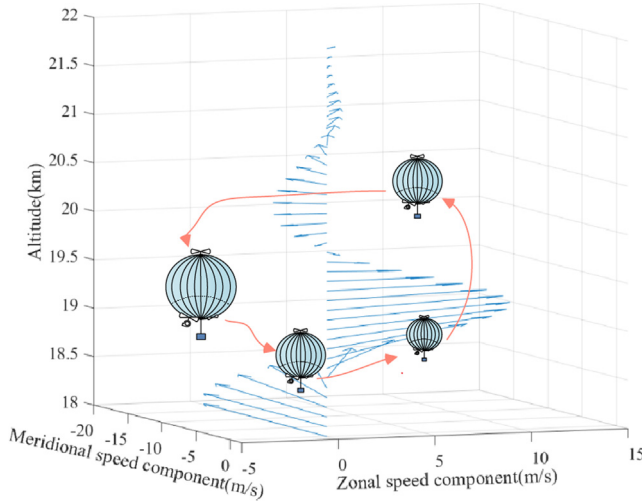


Fig. 4. The principle of balloon station keeping control method.

loon (Pande and Verstraete, 2018; Ramesh et al., 2018), which indirectly leads to the internal pressure exceeding or blowing the specified range (Garg and Emami, 2019). The thermal characteristics of the balloon consist of three parts: helium, air, and film. In addition to the influence of radiation from the external environment, heat transfer among the three parts is also proceeding. The thermal radiation from the external environment mainly includes direct solar radiation q_d , scattered solar radiation q_s , planet albedo solar radiation q_{AP} , sky infrared radiation q_{IR_sky} , ground infrared radiation q_{IR_P} . The internal thermal radiation mainly includes infrared radiation emitted by film q_{IR_film} , convective heat transfer between the external atmosphere and film $q_{ConvExt}$, convective heat transfer between the film and Helium q_{f_He} , convective heat transfer between film and air q_{f_air} , the heat transfer between Helium and air q_{air_He} (Zhang and Liu, 2015). According to the first law of thermodynamics, the temperature differential equations of film, helium and air are expressed by (Yang, 2016).

$$m_f c_f \frac{dT_f}{dt} = q_d + q_s + q_{AP} + q_{IR_sky} + q_{IR_P} + q_{ConvExt} - q_{f_air} - q_{f_He} - q_{IR_film} \quad (1)$$

$$m_{He} c_{v,He} \frac{dT_{He}}{dt} = q_{f_He} + q_{air_He} + \frac{RT_{He}}{M_{He}} \cdot \frac{dm_{He}}{dt} - P_{He} \frac{dV_{He}}{dt} + c_{v,He} T_{He} \frac{dm_{He}}{dt} \quad (2)$$

$$m_{air} c_{v,air} \frac{dT_{air}}{dt} = q_{f_air} - q_{air_He} + \frac{RT_{air}}{M_{air}} \cdot \frac{dm_{air}}{dt} - P_{air} \times \frac{dV_{air}}{dt} + c_{v,air} T_{air} \frac{dm_{air}}{dt} \quad (3)$$

where T_f is the film temperature, $c_{v,He(air)}$, $T_{He(air)}$, $V_{He(air)}$, $P_{He(air)}$, $M_{He(air)}$ are specific heat of helium (air) gas at constant volume, helium (air) temperature, gas volume,

pressure of helium (air) and helium (air) gas mass respectively.

The external heat radiations are given by (Farley, 2005).

$$q_d = \alpha \cdot A_{project} \cdot I_{sun} \cdot [1 + \tau(1 + r_{eff})] \quad (4)$$

$$q_s = \alpha \cdot A_{surf} \cdot I_{sun_s} \cdot [1 + \tau(1 + r_{eff})] \quad (5)$$

$$q_{AP} = \alpha \cdot A_{surf} \cdot I_{albedo} \cdot F_{view} \cdot [1 + \tau(1 + r_{eff})] \quad (6)$$

$$q_{IR_sky} = \alpha_{IR} \cdot A_{surf} \cdot I_{IR_sky} \cdot (1 - F_{view}) \cdot [1 + \tau_{IR}(1 + r_{eff_IR})] \quad (7)$$

$$q_{IR_P} = \alpha_{IR} \cdot A_{surf} \cdot I_{IR_ground} \cdot F_{view} \cdot [1 + \tau_{IR}(1 + r_{IR_eff})] \quad (8)$$

$$q_{ConvExt} = hc_{exter} \cdot A_{surf} \cdot (T_{env} - T_f) \quad (9)$$

where α , τ and r_{eff} are the average absorptivity, transmittance and reflectivity of film, α_{IR} , τ_{IR} and r_{IR_eff} are the infrared absorptivity, transmittance and reflectivity of film. A_{surf} is the surface area of the balloon, and $A_{project}$ is the top projected area. F_{view} is the balloon surface area diffuse-radiant view factor of the planet. hc_{exter} is the external convection heat coefficient.

The internal heat radiations are given by (Zhu et al., 2019).

$$q_{IR_film} = \sigma \cdot \varepsilon \cdot \alpha_{IR} \cdot A_{surf} \cdot T_f^4 \cdot (1 + r_{IR_eff}) - 2 \cdot \sigma \cdot \alpha_{IR} \cdot A_{surf} \cdot T_f^4 \quad (10)$$

$$q_{f_He} = H_{free}^{He} \cdot A_{surf} \cdot (T_f - T_{He}) \quad (11)$$

$$q_{f_air} = H_{free}^{air} \cdot A_{surf} \cdot (T_f - T_{He}) \quad (12)$$

$$q_{air_He} = K \cdot A_{mem} \cdot (T_{air} - T_{He}) \quad (13)$$

where H_{free}^{He} and H_{free}^{air} are the helium free convection heat coefficient and air free convection heat coefficient respectively. A_{mem} and K are the film area and heat-transfer coefficient of the diaphragm separating air and helium.

2.3. Dynamic model

The buoyancy and gravity of the balloon in the vertical direction can be expressed as.

$$B = \rho_{atm} \cdot g \cdot V \quad (14)$$

$$G = m_{total} \cdot g \quad (15)$$

$$m_{total} = m_{He} + m_{air} + m_{film} + m_{mem} + m_{payload} \quad (16)$$

where the parameter m_{total} is the total mass of the balloon system as the Eq.(16) shows and ρ_{atm} is the atmosphere density of current altitude, and the gravity of the balloon can be deposited into Helium mass, air mass, film mass, membrane mass and payload mass.

The force that drives the balloon in horizontal direction is provided by the horizontal wind, which can be expressed by.

$$D = 0.5 \cdot \rho_{atm} \cdot v_r^2 \cdot C_d \cdot S \quad (17)$$

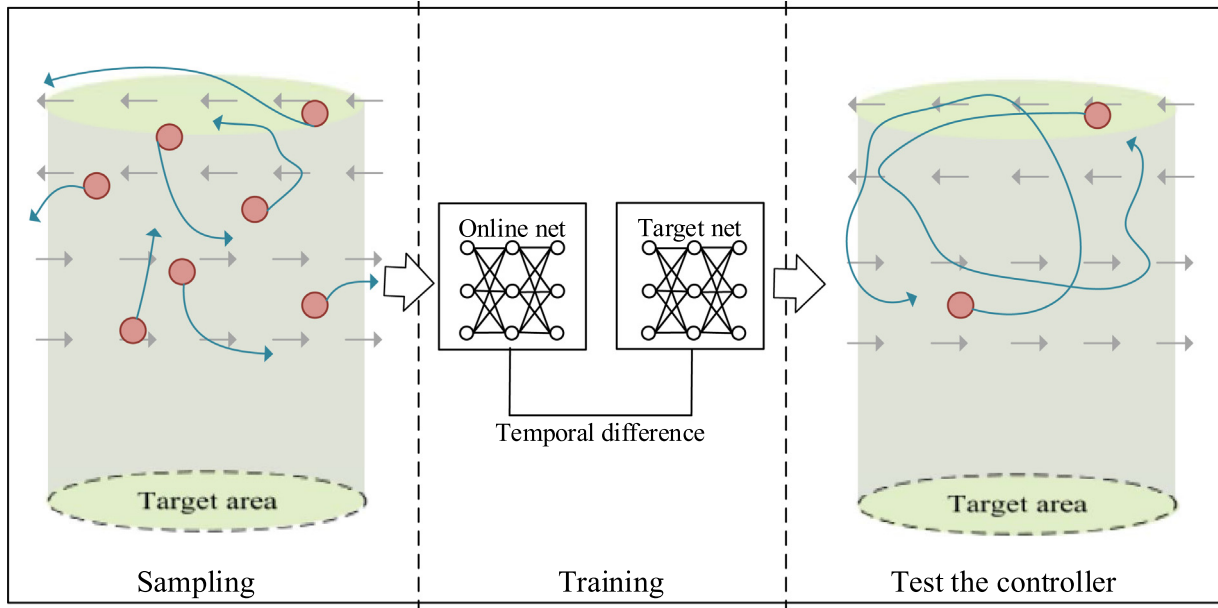


Fig. 5. Reinforcement learning and training process.

where C_d is the drag coefficient. S is the windward reference area, which is described as $S = V^{2/3}$. In the North-East-Down Coordinate System, v_r is composed of the differences between the component of wind speed (v_{wx}, v_{wy}, v_{wz}) and balloon speed (v_x, v_y, v_z) on each coordinate axis correspondingly.

$$\begin{cases} v_r = \sqrt{v_{rx}^2 + v_{ry}^2 + v_{rz}^2} \\ v_{rx} = v_{wx} - v_x \\ v_{ry} = v_{wy} - v_y \\ v_{rz} = v_{wz} - v_z \end{cases} \quad (18)$$

Thus, the drag force component D_x , D_y and D_z can be expressed by.

$$\begin{cases} D_x = D \cdot v_{rx} / v_r \\ D_y = D \cdot v_{ry} / v_r \\ D_z = D \cdot v_{rz} / v_r \end{cases} \quad (19)$$

To sum up, the differential equations of balloon displacement and velocity are as follows.

$$\begin{cases} \dot{x} = v_x \\ \dot{y} = v_y \\ \dot{z} = v_z \\ \dot{v}_x = D_x / (m_{total} + m_{add}) \\ \dot{v}_y = D_y / (m_{total} + m_{add}) \\ \dot{v}_z = (B + D_z - m_{total} \cdot g) / (m_{total} + m_{add}) \end{cases} \quad (20)$$

where m_{add} is a virtual mass defined as Eq.(21), where the coefficient is $C_V = 0.5$ (Anderson and Taback, 2012).

$$m_{add} = C_V \cdot \rho_{atm} \cdot V \quad (21)$$

3. Station-keeping control strategy by HVS-PER-D3QN

To ensure the effectiveness of the control strategy, it is important to create a training environment that accurately simulates the atmospheric environment (including wind field and temperature). Combined with the environment model and the RL framework, the trajectory of the balloon with different control strategies can be simulated. Since the balloon is an agent moving in continuous space, it may appear in various positions. The action space is defined according to the three modes of control altitude, including inflation, deflation and closed. Therefore, discrete actions of the executive mechanism are adopted in this paper. It is implausible to use a table to depict the balloon's continuous motion state space, but by combining dueling double Q-learning and deep network (D3QN), the 'curse of dimensionality' can be effectively solved. Dueling double deep Q-learning is a form of reinforcement learning in which a neural network is used to estimate the value of the Q-function. It performs better in terms of convergence and stability than a fully linked neural network. The agent is capable of learning a strategy in continuous state space without previous information, as the neural network is effectively a transform between two vectors of arbitrary dimension (Rao, 2000). The nodes in the output layer correspond to the Q value estimation of different actions, which will be the reference for the controller to select actions.

Fig. 5 displays the entire process of developing a control plan. The entire procedure is divided into three stages: sampling, training, and controller performance evaluation. First, the sampling phase simulates the random release of a large number of balloons into the environment to collect environmental samples and flight data. Secondly, the neural network will use historical samples to seek control strate-

Table 1
State parameters setting.

Feature	Range	Notes
East position s_x	–50 km ~ 50 km	Subject to $\sqrt{x^2 + y^2} \leq 50$
North position s_y	–50 km ~ 50 km	
Air mass s_{ma}	7.41 kg ~ 92.9 kg	In accordance with specific conditions
Temperature s_{Ta}	–	
Bearing angle s_{angle}	–180° ~ 180°	s_{angle} encodes the bearing of winds at altitude ranging from ± 1 km
Wind speed s_{ws}	–	In accordance with specific conditions

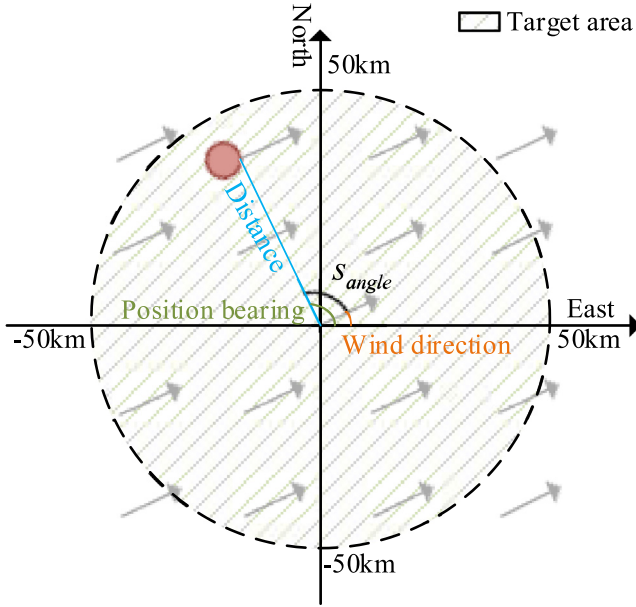


Fig. 6. Schematic diagram of state parameters (Top view of the 3D space).

gies that maximize the total reward based on the current reward function. Thirdly, the networks developed during training are evaluated for starting condition sensitivity, and the networks with the best performance are chosen as the balloon's primary controller.

3.1. Markov decision process model

Q-learning and extension methods are based on the Markov decision process (MDP), which is a discrete, sequential decision problem for a stochastic environment (Kruse et al., 2019). The Q-learning method based on MDP can be described by five elements $[S, A, R, P_{ss'}^a, Q(s, a)]$. The objective of reinforcement learning algorithm is to find an policy π^* that maximizes the expected sum of the discounted rewards.

$$\begin{aligned}
 Q_{\pi}(s, a) &= E_{\pi}[r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} + \dots | S_t = s, A_t = a] \\
 &= E_{\pi}[r_{t+1} + \gamma Q_{\pi}(s_{t+1}, a_{t+1}) | S_t = s, A_t = a]
 \end{aligned}
 \quad (22)$$

In this paper, the set of each element of MDP based on balloon training environment is as follows.

Table 2
Action parameters setting.

Action space	Action	Air mass change rate
a_1	Deflation	–0.006 g/s
a_2	Closed	0 g/s
a_3	Inflation	0.004 g/s

3.2. State space S

In the flight process, the state is a vector composed of location coordinates, thermal conditions, and the atmospheric environment. The position coordinate $[s_x, s_y, s_z]$ determines the attainment of successful station keeping. Air mass s_{ma} and air temperature s_{Ta} are related to the over-pressure of the balloon, which is crucial to the safety of the balloon. The specific parameters and value range of states are set as shown in Table 1. Fig. 6 presents the source and calculation method of these parameters.

3.3. Action space A

According to the design of the altitude control system, the action space of the actuator is composed of discrete actions of the compressor, and the corresponding action space $A = [a_1, a_2, a_3]$. The meaning of action space parameters is shown in Table 2.

3.4. Transition probability $P_{ss'}^a$

The states transition of the balloon are resolved based on the balloon theoretical model, so the transition probability $P_{ss'}^a = 1$ for each action.

3.5. Reward function R

Reward function is an important part of Q-learning and it should reflect the station-keeping purpose of balloon. The reward function proposed in this paper is defined as the combination of three parts as shown in Eq.(23).

$$R(s_t) = N(\omega\theta - \omega_0 l) + bias \quad (23)$$

where l is the distance between the balloon and the origin point as shown in Fig. 6. Because the wind is beneficial for the balloon to return to the origin point with the increase of the angle θ . The angle θ is the angle between the position

bearing of the balloon and the wind direction, which also equals the parameter s_{angle} . Factor ω_0 and ω are positive. The parameter N ensures that the range of the sum of $\omega_0 l$ and $\omega \theta$ does not change with the weight, and the parameters ω_0 and ω are used to adjust the weight of distance factor l and angle factor θ , which aims to find the reward function with optimal weight distribution. Moreover, $bias$ is added to distinguish the inflation/deflation from the closed action, because the working time of the compressor should be infrequent to conserve energy. When the balloon is close to the center point P_0 or the angle between wind direction and current trace point azimuth is large, it can achieve a better station keeping performance.

3.6. Q-learning and action value function $Q(s_t, a_t | \theta)$

Q-learning is based on the maximum value of $Q(s_t, a_t | \theta)$ in current state-action space to select action strategies rather than directly searching the policy. The parameters θ presents the network parameters, and the agent will select the optimal action according to the maximum value of the output layers. To solve the problem of maximum value of $Q(s_t, a_t | \theta)$, Temporal Difference (TD) methods are used to attempt to find the optimal Q value. By using TD(0) method, the agent samples in the unknown environment and save the samples which are consist of current states s_t , next time step state s_{t+1} and real-time reward r_{t+1} (Daskiran et al., 2017). Based on the TD(0) method, the error between the predicted value of target network $Q(s_t, a_t | \theta)$ and online network $Q(s_t, a_t | \theta^-)$ is used to update the parameter θ^- by gradient descent. Combined with the deep neural networks and gradient descent method, the recursive equation is as follow:

$$Q(s_t, a_t | \theta) = \begin{cases} r_t & \text{if episode terminates at next step} \\ r_t + \gamma \max_{a_{t+1}} Q(s_{t+1}, a_{t+1} | \theta) & \text{otherwise} \end{cases} \quad (24)$$

$$L(\theta^-) = \left[(Q(s_t, a_t | \theta^-) - Q(s_t, a_t | \theta))^2 \right]$$

3.7. Sampling and High-Value samples Prioritized Experience replay

The TD method samples the unknown environment based on time step, and its characteristics determine that there is a strong correlation among the samples. In the process of training neural networks for numerical approximation, the correlation of samples can lead to instability. Therefore, a replay buffer is needed to store past experience samples before the training process so as to sample randomly from the buffer during training, which can eliminate

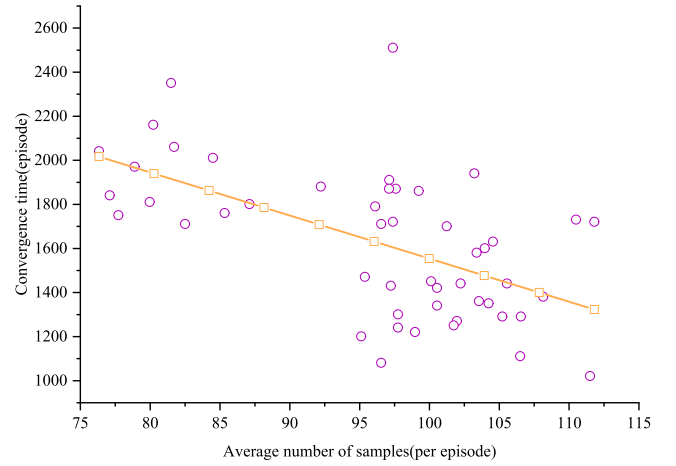


Fig. 7. Relationship between convergence time and average number of samples per episode.

the time correlation (Schaul et al., 2015). The episode ends when the balloon leaves the target area, which is $\sqrt{s_x^2 + s_y^2} > l_{\max}$, where $l_{\max}$ is defined as the maximum effective working radius, or the overpressure Δp is out of the specified range $[\Delta p_{\min}, \Delta p_{\max}]$. Thus, the replay buffer needs to consume hundreds of episodes to be filled. The approximation accuracy will increase as the sample number increases. Considering the computational cost and training effect, the capacity of the replay buffer is designed as 96,000 samples in this paper.

The object of the high-altitude balloon in this paper is to stay within the target area as long as possible, so the long-term station-keeping samples generated in each episode are more valuable for training. It is expected that the value

of samples in the buffer increases gradually during the process. In Fig. 7, these points present the relationship between average number of samples in each episode and time cost of subsequent training. The larger average value means the fewer episodes it costs to fill the replay buffer, and the sample experiences are more profitable to guide the balloon to realize long-term station keeping. The least square method is used to fit these points, and the result indicates that the long-term samples can speed up the training process.

The samples in the replay buffer will be covered indiscriminately during the training process, no matter these

samples have positive effect on the training or not. Prioritized Experience Replay evaluates the priority of all samples depending on the value function, which makes the high priority samples more likely to be selected. It requires that all samples in the replay buffer are traversed to update their priority when new samples are added. At the start of the training, the strategy is set to be random, and each step is rewarded differently. Among the large number of sam-

the current state. Since blindly choosing the action with the maximum value often leads to overestimation, the alternative renewal double networks are used for the reference and prediction of Q value. The online network updates its parameters in real time with the increasing episode number, but the frequency of target network parameter updates is slower. The iterative formula of the target and online networks is:

$$Q(s_t, a_t | \theta) = \begin{cases} r_t & \text{if episode terminates at next step} \\ r_t + \gamma \max_{a_{t+1}} Q(s_{t+1}, \arg \max_{a_{t+1}} Q(s_{t+1}, a_{t+1} | \theta^-) | \theta) & \text{otherwise} \end{cases} \quad (25)$$

$$L(\theta^-) = \left[(Q(s_t, a_t | \theta^-) - Q(s_t, a_t | \theta))^2 \right]$$

ples, the high-value sample indicates that balloons could get larger rewards under the series of action. These examples contribute to strategy development, and they should not be prematurely overwritten.

As the replay buffer in Fig. 8, the original replay buffer is set up as two partitional parts: the red part for storing the normal samples and the blue part for storing the High-Value Samples. The High-Value Samples are stored in the blue specific part in the buffer, and they are not covered until higher value samples are generated. Fig. 8 presents the details of the Prioritized Experience Reply based on the High-Value Samples (HVS-PER). At the beginning of training, the sampling rule is completely random, and the identification standard for the ‘High-Value Samples’ is set at a low level. The total value of generated samples has to be above the identification standard before they are admitted to the buffer, and the cache is prepared for storing the samples before the total value is assessed. With the increase of training, the total value of some High-Value Samples will be stored in the blue part, and the sampling probability of these High-Value Samples will be increased. Meanwhile, the identification standard for admission to store High-Value Samples will be increased with the total value increasing, which can help improve sample quality and realize the effective use of High-Value Samples. Because the storage address of high-value samples is known during the sampling phase, it is convenient to select High-Value Samples for preferred prioritized replay without traversing the full buffer. The HSV-HVS-PER method does not affect the convergence of the training process.

3.8. Dueling double deep Q learning network framework

This study takes the dueling network as an approximate function to estimate $Q(s_t, a_t | \theta)$. The value of $Q(s_t, a_t | \theta)$ presents the total reward under different action selections at

After testing, the full connection network performs best, which has one input layer, five hidden layers, one layer set as the advantage function, and one output layer. The input layer has 11 neurons, which correspond to the 11 set states. The output layer has 3 neurons, which correspond to three actions of actuators.

In this study, the number of neurons is determined by balancing the training effect and the computational cost. In particular, the generalization effect of deep reinforcement learning is not necessarily better with the increase of network depth. The next state is updated by taking the control frequency as a cycle. The derived states-rewards sequence stored in the buffer reflects the concerns of the balloon station keeping effect in the real wind field. As in manned flight, decisions are primarily based on the relative position and wind direction. With the increase of training time, the decision-making method can help the neural network converge and maximize the Q function. The network is fixed to be stable and can be used as the main controller to make flight decisions to realize long-term station keeping.

4. Simulation results and discussion

4.1. Balloon altitude control system performance

Taking a stratospheric balloon with a diameter of 15 m as an example, the main design parameters of the balloon are provided in Table 3. The upper limit h_{max} and lower limit h_{min} of balloon flight for this scenario are about 18–22 km. A balloon agent is built with the parameters in the HVS-PER-D3QN reinforcement learning framework to simulate the flight effect of the controller and evaluate the performance of the controller.

To validate the rationality of the design and eliminate the potential hazards of inflation and exhaust and complex

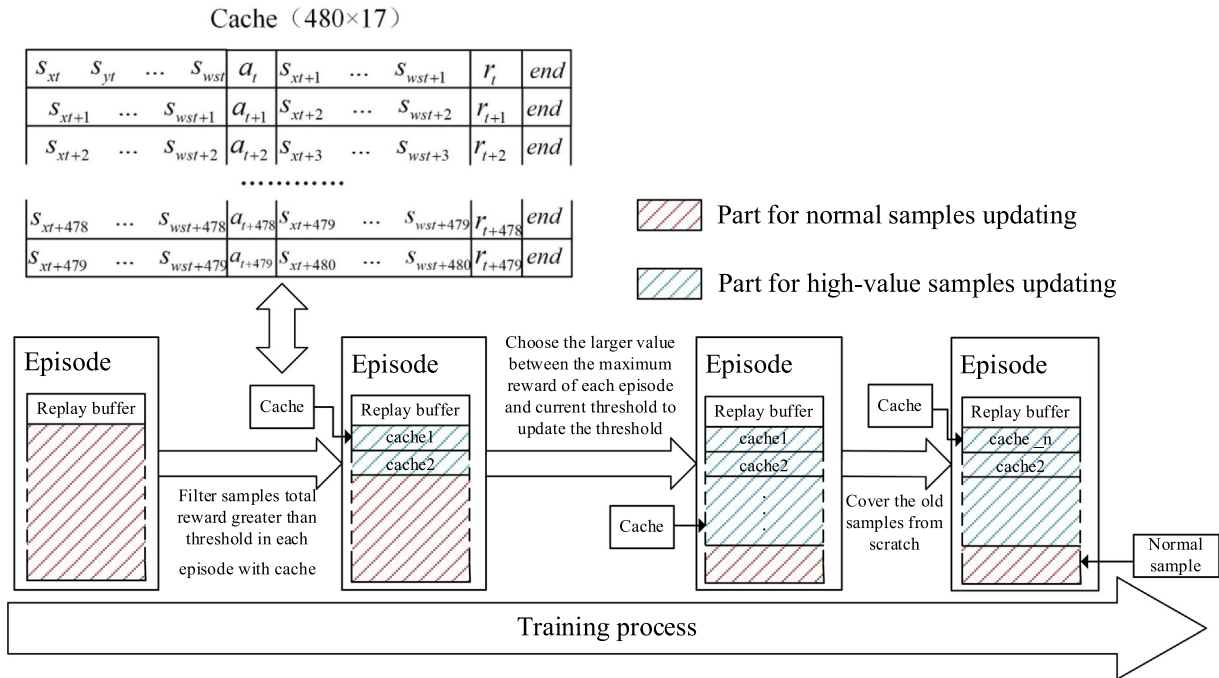


Fig. 8. Schematic diagram for the process of High-Value Samples Prioritized Experience Replay.

Table 3
Design parameters of high-altitude super-pressure balloon.

Parameters	Value
Diameter	15 m
Helium mass	16.8 kg
Film material density	0.0722 kg/m ²
Minimum air ballonnet volume	102 m ³
Payload	36 kg
Maximum safe differential pressure	1200 Pa (Du et al., 2019b)

thermal radiation on the safe flight of the balloon, the temperature and internal pressure changes of the balloon within 24 h are calculated. Fig. 9 represents the temperature change process in the balloon caused by the complex thermal radiation of the atmosphere. Without the effects of inflation and deflation, the maximum temperature appears at 12:30 pm with a value of 246 K. Then the temperatures decrease with the setting of the sun and drop rapidly to 220 K after sunset. The maximum difference in temperature within 24 h is about 26 K.

According to the overall design, the balloon can adjust altitude by increasing or decreasing the air mass. The minimum air mass is determined by the minimum air ballonnet volume, and the ideal maximum air mass is determined by the expected maximum altitude adjustment range. However, the influence of the temperature is complex, and the rise in temperature may cause the balloon to burst. The overpressure is calculated by.

$$\Delta p = p_{\text{internal}} - p_{\text{atm}} \quad (26)$$

where p_{internal} is the pressure of the internal gas. p_{atm} is the atmosphere pressure at current altitude.

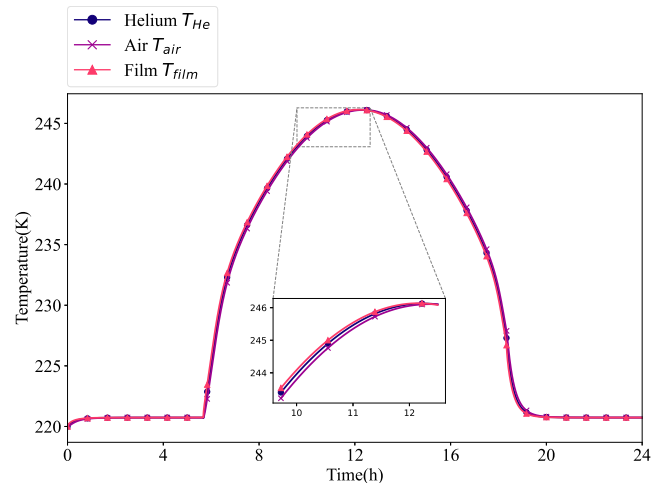


Fig. 9. Variations of the temperature of air and Helium and film.

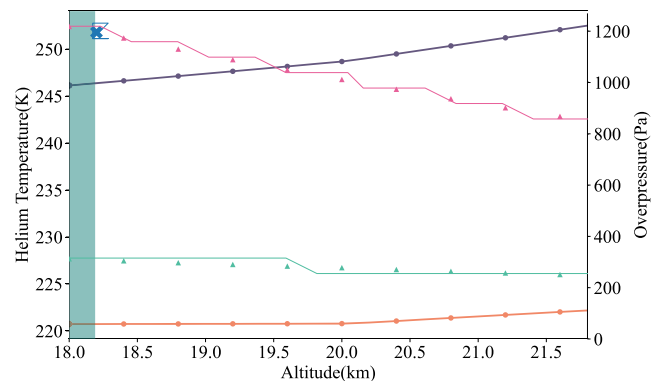


Fig. 10. Variations of gas temperature and overpressure within 24 h at different altitude.

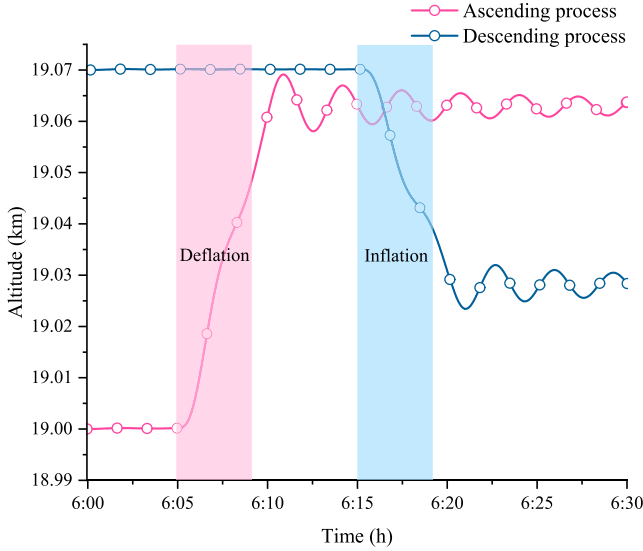


Fig. 11. Variations of altitude during the venting and pumping process.

To ensure the safety of the balloon, the variation of the pressure difference between the inside and outside of the balloon under the influence of gas temperature and mass is analyzed. Fig. 10 shows the extreme values of temperature and overpressure during the 24 h with different floating altitudes. Because the maximum overpressure that film material can endure is $\Delta p_{max} = 1200$ Pa (Table 3), the result indicates that the balloon cannot keep floating below 18.2 km for 24 h. The green shadow part presents that the maximum overpressure exceeds Δp_{max} when the temperature of the internal gas reaches maximum value. Therefore, the target altitude of the ACS is limited by temperature and differential pressure, which should be used as independent variables of the controller as comprehensively as possible.

According to the controller order, the ACS can exhaust or pump gas from the air bag. Due to the balloon's motion characteristics, it cannot respond rapidly to high frequency orders. Thus, it is not necessary to adjust the altitude frequently. We take three minutes as a period to send the control commands. Fig. 11 illustrates the ability of altitude adjustment. During the descent process, the balloon is filled with 0.72 kg of air at a rate of 4 g/s in three minutes, lowering the height by 35 m. And the balloon exhaust 1.08 kg of air at a rate of 6 g/s in three minutes, raising the balloon's height by around 65 m. The drop height between the different wind layers with different wind directions in the quasi-zero wind layer is generally hundreds of meters. Therefore, the resolution of the control step is sufficient to allow the balloon to fly back and forth between distinct wind layers.

4.2. Simulation model and environment development

The RL approach is built on a control strategy that involves the training environment, state, and the balloon's

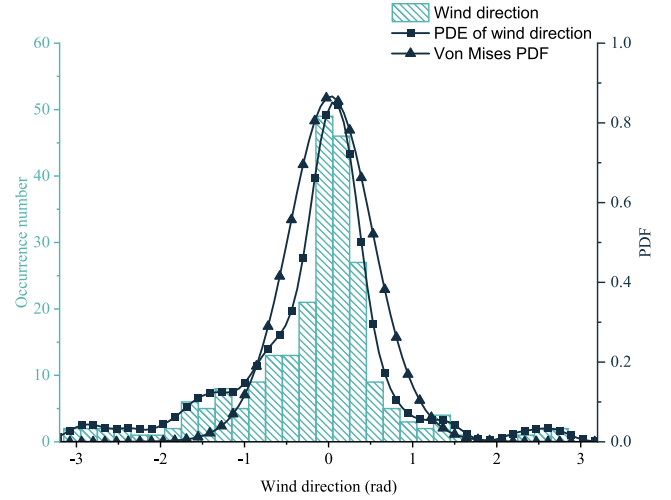


Fig. 12. Occurrence number and probability density of wind direction at 22 km.

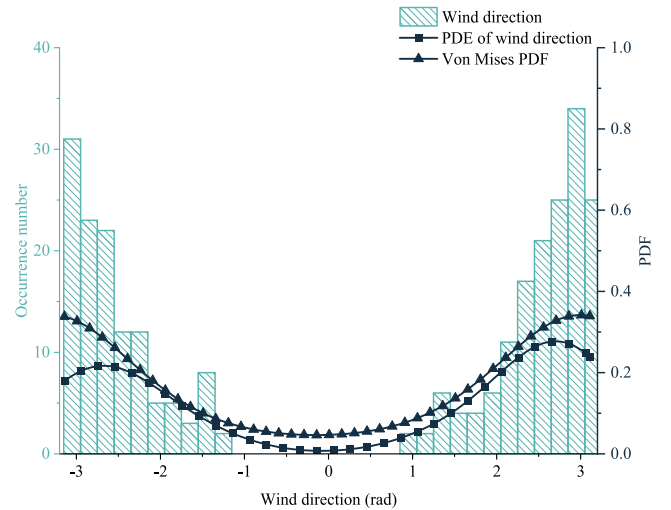


Fig. 13. Occurrence number and probability density of wind direction at 18 km.

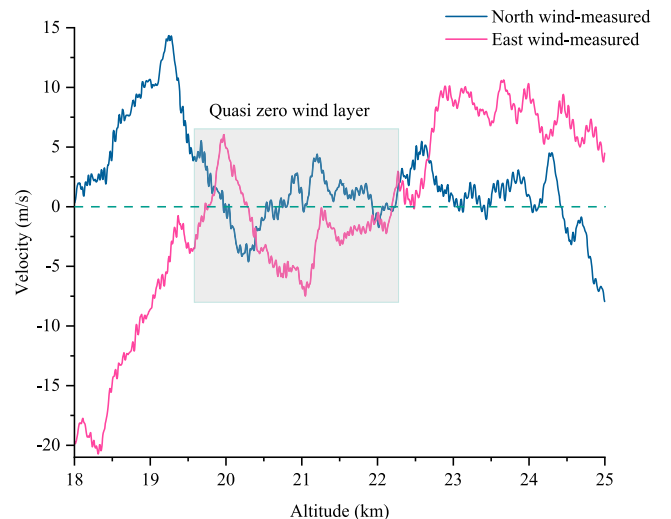


Fig. 14. Different altitudes wind field test results.

$$\begin{aligned}
s_{x0} &\in [-10\text{km}, 10\text{km}] \\
s_{y0} &\in [-10\text{km}, 10\text{km}] \\
s_{z0} &\in [18.5\text{km}, 20.5\text{km}]
\end{aligned} \quad (29)$$

To ensure that all the parameters have the same order of magnitude, the states need to be kept at a same level during the training process.

$$s_t = \left[\frac{s_x}{20} \& \frac{s_y}{20} \& \frac{s_z-18.2}{3.6} \& \frac{s_{ma}}{92.9} \& \frac{s_{ra}-230}{26} \& \frac{s_{angle}}{180} \& \frac{s_{us}}{20} \right] \quad (30)$$

By applying Eq. to the states in input layer, the output layer can calculate the value of different actions in current states.

Table 5 presents the episode termination conditions, which are formulated according to the specific flight. In this paper, the overpressure of the balloon is kept between Δp_{min} and Δp_{max} . And the balloon is expected to fly over a circle with a radius of 50 km, and the center of the circle is defined as the center point P_0 . The effective working time t_{eff} is defined as the period of flight in the circular region.

4.3. Station-keeping strategy learning process and results discussion

The reward function is defined as a distance-based and wind-direction based function Eq.. According to the balloon's station-keeping flight characteristics, the wind-direction component is a direct element in modifying the balloon's flight direction, which can quickly alter the balloon's trajectory. While the inflation and deflation of the balloon cannot directly affect the displacement of the balloon in the horizontal direction, the distance-based component is a decisive factor in the achievement of successful station keeping. As a result, the impact of reward function composition is investigated under various weight proportions. These two factors weights are set as $\omega/\omega_0 = [0.1, 1, 2, 3, 4 \cdots 13, 14, 15]$. To quantify the effect of these weights, all balloons with varying reward functions are placed in the same simulated environment.

By training the network with a variety of different reward functions, it is possible to adjust FCNN to a variety of distinct preferences. To assess the link between wind-direction and distance-based factors, all FCNNs are tested for their ability to control the balloon while remaining in the target position. Within 24 h, the entire flight track has 480 trace points as a result of sampling every three minutes.

Based on the episode termination condition shown in Chapter 4.2, the flight route and trace point distributions are examined and shown in Fig. 16. The 0 km–30 km seg-

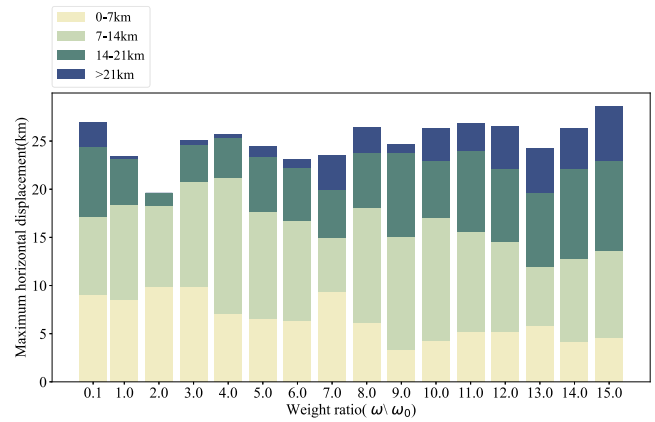


Fig. 16. Variation of the maximum horizontal displacement and the distribution of trace points with the increase of the weight ratio.

ment is separated into four portions: 0 km–7 km, 7 km–14 km, 14 km–21 km, and greater than 21 km. These divisions are utilized to aid statistics in determining the distribution of trace points locations within each section. Fig. 16 presents that the longest horizontal distance reduces as the weight ratio ω/ω_0 grows when it is less than 2.0, but increases when it is more than 2.0. When $\omega/\omega_0 = 2.0$, the flight radius of the balloon is 19.6 km and the balloon's trace points distribution is plainly closest to the center point P_0 . Additionally, the varied colors in Fig. 16 reflect the percentage of time the balloon appears at various positions throughout the flight, and the darker color areas imply the balloon is further away from the central point. As a result, the balloon can be as near to the center as feasible if the ratio $\omega/\omega_0 = 2.0$ is used.

Referring to the distribution of trace points shown in Fig. 17, 16 networks with a better station keeping effect are chosen from the multiple networks trained with each different reward function, and the horizontal projections of the motion simulation trajectories are recorded. The density of these points may be computed and represented in color using the Gaussian kernel density estimation technique. As can be seen, when different weight ratios are applied, the distribution of sampling points in the trajectory with a larger radius is sparse, whereas the distribution of sampling points in the trajectory with a smaller radius is more concentrated, implying a higher degree of coincidence between balloon flight paths during different periods. Since the balloon travels across distinct wind layers, the control technique should converge on a stable and repeatable plan, while the small radius makes the flying process more fault-tolerant. Fig. 18 depicts the training procedure, which demonstrates that the balloon's maximum reward can reach approximately 140. 'Early stop' can avoid the decrease in success rate caused by overfitting, which means that training can be stopped when the success rate stays steady.

Fig. 17 Variation of kernel density estimation of horizontal trace point with different reward function.

Table 5
Episode termination conditions.

Restrictions	Value
l_{max}	50 km
Δp_{max}	1200 Pa
Δp_{min}	0 Pa

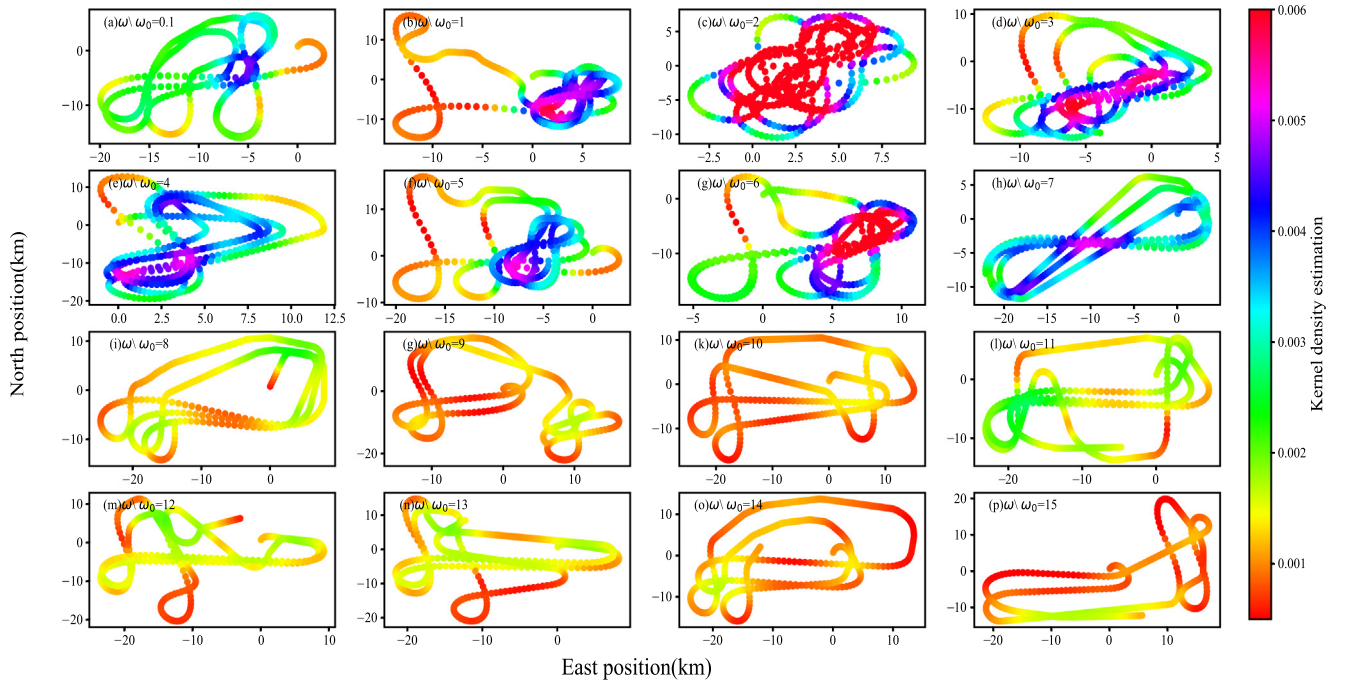


Fig. 17. Variation of kernel density estimation of horizontal trace point with different reward function.

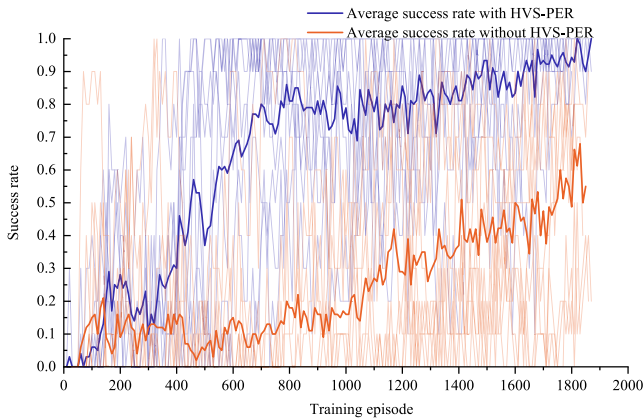


Fig. 18. Experimental results of success rates with different methods.

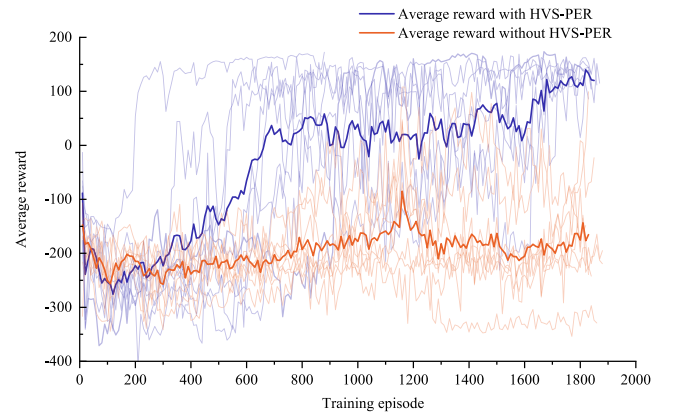


Fig. 19. Variation of average rewards during training process.

The effect of the HVS-PER method is shown through several experiments. These results demonstrate the superiority of the HVS-PER over the random sampling method. Fig. 18 presents that the success rates increase as the number of training episodes increases, and the HVS-PER method based on high-value samples can significantly improve the training speed of dueling networks. It is because the screened high-value samples can be safeguarded during the training process. It is also considered the performance of the optimal solution obtained by these two methods. What is required is that the flight radius is kept as short as possible during the whole day, and the reward value also serve as a descriptor for the flight distance. Fig. 19 illustrates the balloon agent's rewards over a 24-hour period, which is used to evaluate the trajectory strategy. Multiple experiments indicate that networks con-

structed using the HVS-PER method outperform networks constructed using the random sampling method. This implies that the method may direct the balloon to remain inside a smaller region, and the HVS-PER technique also enhances the capability of the training process to escape from the local optimum.

After the sampling phase, the strategy converges and remains stable after 1800 episodes. Owing to the training record, the training process usually lasts about 35 min. On the 3.3 Ghz processor, the algorithm takes about 10 ms to select the action according to the output of the network. The time of the solution process would increase as the processor frequency decreases. In this work, the balloon updates the states and actions every three minutes, so the algorithm run time is not an issue.

4.4. Station-keeping strategy performance analysis

The station-keeping strategy's performance is determined by the average horizontal displacement d_{avg} , the sensitivity of initial position P_0 , and the effective working time t_{eff} . The effective working time t_{eff} is defined as.

$$t_{eff} = \frac{24 - \text{time}_{d>50}}{24} \times 100\% \quad (31)$$

Firstly, the performance of the network is tested in the environment with steady wind, and the balloon trajectory in 24 h is an important indicator of evaluation. Fig. 20 presents the trajectory and the radius of the level trajectory in 24 h. Clearly, the balloon can keep the station within about 10 km of the constant wind environment.

The inner gas temperature and differential pressure are illustrated in Fig. 21. Inflation or deflation influences the gas temperature and the differential pressure is subject to the temperature changes. Helium has a maximum temperature of about 252 K at noon, while air has a temperature of about 255 K. After sunset, the minimum temperatures of helium and air are 215 K and 212 K. Meanwhile, the differential pressure maximum and minimum values are 1081 Pa and 115 Pa, respectively. Thus, the balloon can maintain the design shape for up to 24 h safely.

Previous studies (Du et al., 2019b) have suggested that the balloon can search the wind layer according to the maximum degree of bearing between the position and the wind. The performance of the controller (Balloon B) is compared with the method of previous studies (Balloon A), and the result is shown in Fig. 22. At the adjustment stage, both balloons are blown far off the center point even leaving the region, and the balloons are back to the center point at about 10:00. At the stable stage, the average distance of balloon A is 14.58 km and balloon B is 7.62 km. It is obvious the balloon B has the ability to advance

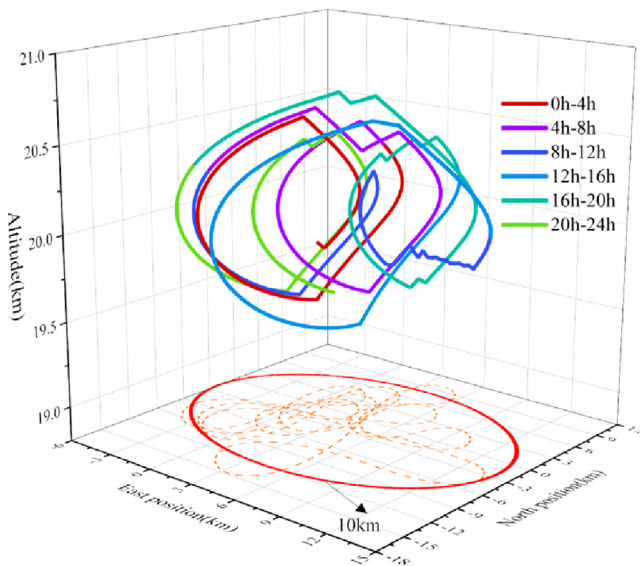


Fig. 20. Trajectory of the balloon with RL method in 24 h.

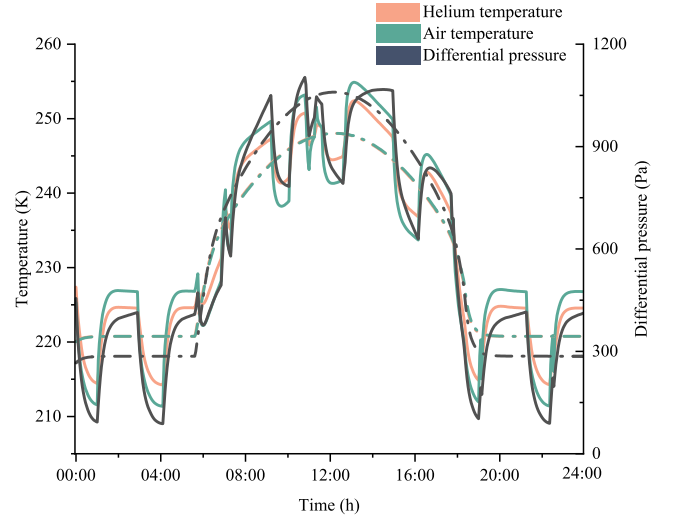


Fig. 21. Variation of the Helium temperature, air temperature and differential pressure.

regulation, because the RL controller can select the optimal wind layer before the balloon starts to move away from the center point.

In order to study the sensitivity of the initial point P_0 , these balloons are put at various locations and the effective working time t_{eff} , average horizontal displacement d are recorded. Considering the disturbance of the unknown wind change, the bandwidth parameter κ is set to 1–8 and ∞ . Where $\kappa=\infty$ means that the wind is steady in the whole day. It contributes to further analysis of the anti-interference ability of these networks.

The average displacement of Balloon A and Balloon B with various beginning positions and bandwidth parameter κ is shown in Fig. 23. The average horizontal displacements of balloons increase as the wind field becomes more variable, and the balloon's horizontal displacement at various initial starting places is counted separately. The average displacements of balloon B deployed at 401 initial points are less than the displacements of balloon A deployed at the same points, and the number of the initial points is 471, 494, and 560, as κ equals 7, 4 and 1 respectively. Because balloon B could use the priority altitude adjustment capability ahead of time, it could reduce flight speed before reaching the center point and ensure that it stayed as close to the center point as possible. Additionally, when the wind field disturbance increases, the station-keeping effects of both balloons decrease at all places. Obviously, when κ equals 1, the station-keeping effect of balloon A is inferior to that of balloon B at all altitudes. At 18 km and 21 km with strong wind speeds, the deployment impact of balloon B with different initial points is the first to diminish, but the deployment at 19–20 km with the quasi-zero wind layer is less affected.

Fig. 24 displays the change of the maximum horizontal displacement d_{max} with different initial points P_0 . With the increase of value κ , the overall value of d_{max} tends downwards at different P_0 . It is because the balloon needs to take

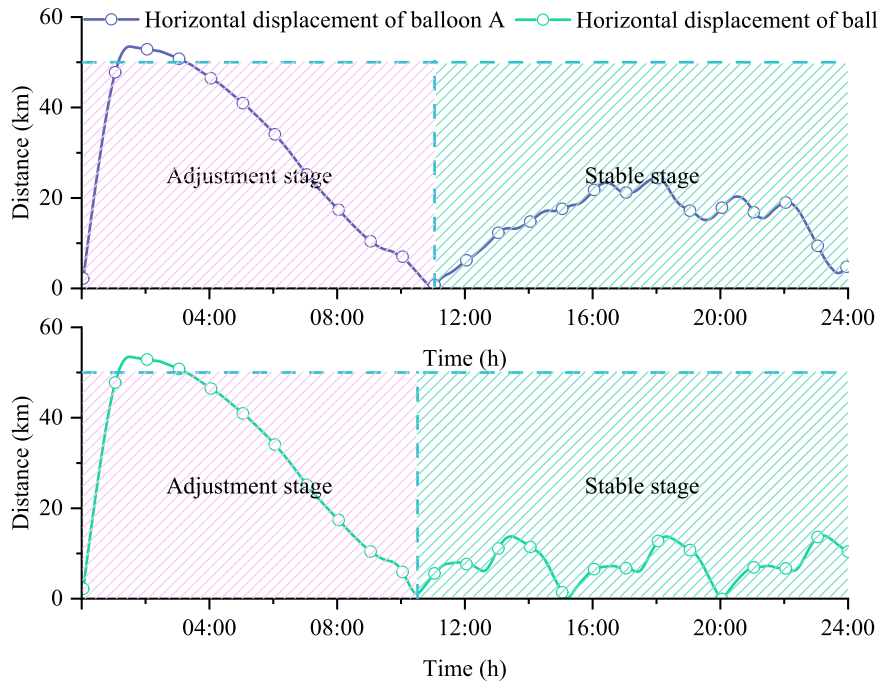
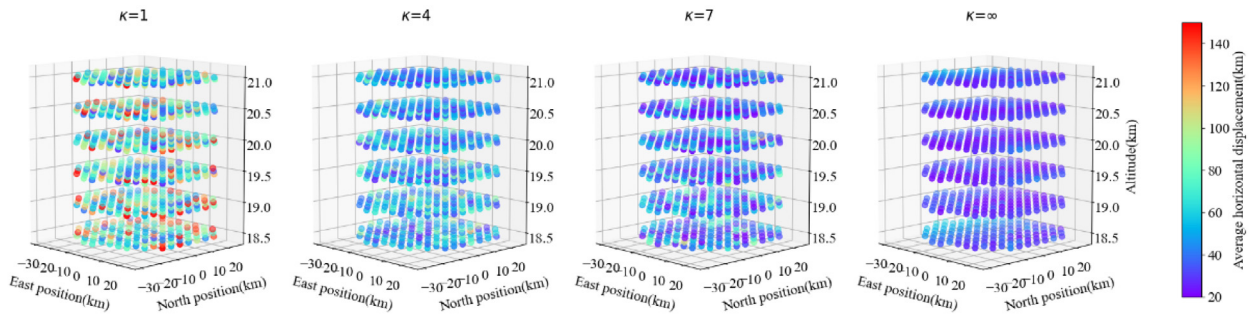
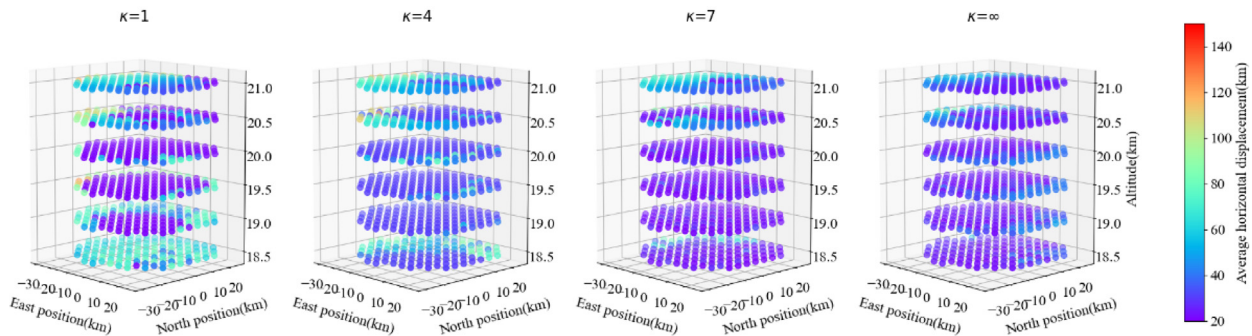


Fig. 22. Comparison of the horizontal displacement within 24 h with different method.



(a) Average horizontal displacement of Balloon A.



(b) Average horizontal displacement of Balloon B.

Fig. 23. Comparison of the average horizontal distance with different initial points.

several minutes before it arrives at the wind layer decided by the controller. The uncertainty of the wind also increases the confusion of the wind field, which causes

the adjustment stage to be longer than before. Especially at about 18.5 km, the wind velocity is greater than at other altitudes. If the wind direction suddenly becomes

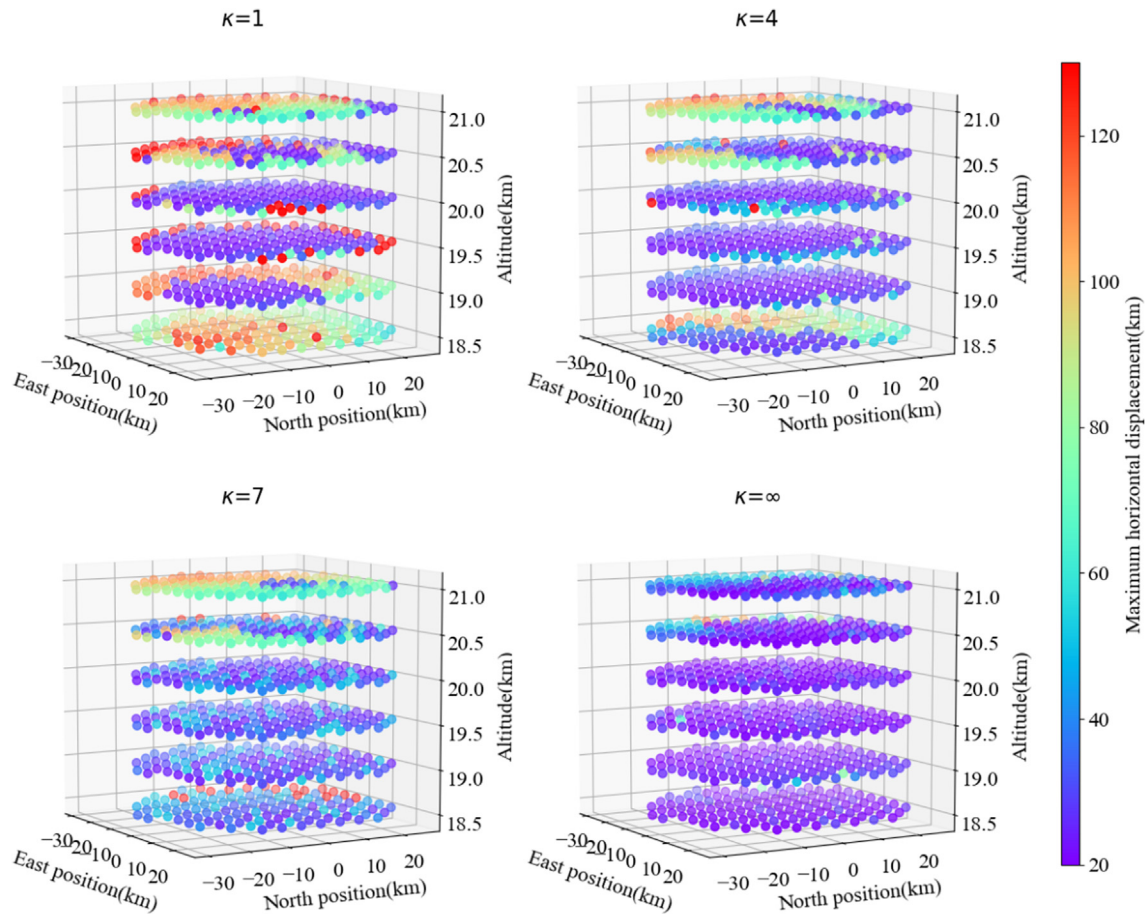


Fig. 24. Maximum horizontal displacement with different initial positions.

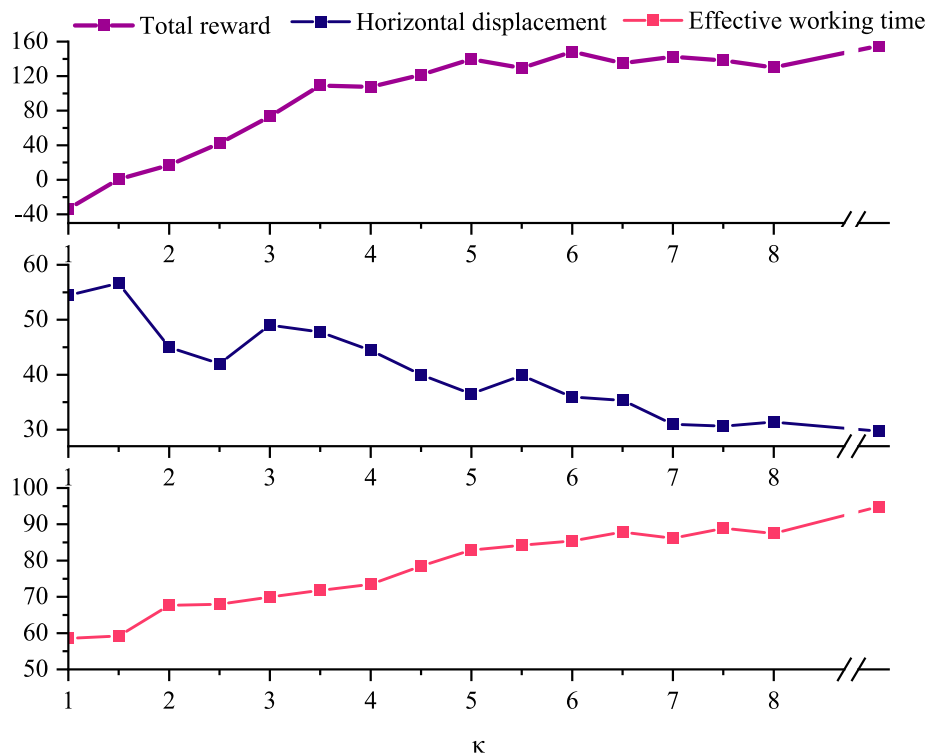


Fig. 25. Variation of total reward, horizontal displacement and effective working time with the increase of bandwidth parameter κ .

unsuitable for approaching the center point, the balloon would fly far away because of the high wind velocity.

Further statistical results of the average total reward, average horizontal displacement d_{avg} and average effective t_{eff} with all tested initial points P_0 are shown in Fig. 25. With the increase of κ , the value of average total reward increases from -40 to 140 , while the effective working time t_{eff} increases from 60% to 90% , and even 100% . And the average horizontal displacement d_{avg} decreases from 55 km to less than 30 km. When the parameter κ increases to 6 , the performance of the controller tends to be stable as it works in a stable wind field. Combined with the local historical and forecast wind data, selecting an appropriate initial position is helpful to keep the station of the balloon in the event of a limited change in wind field.

5. Conclusions

This work proposes the balloon station keeping control strategy based on the HVS-PER-D3QN method with a neural network in a real wind field. The balloon is treated as an agent for Reinforcement Learning, and the optimal action strategy is obtained through optimization of the network. A Prioritized Experience Replay based on the high-value samples is developed to speed up the training process. The anti-disturbance performance of the controller is tested by the balloon total reward, horizontal displacement, and effective working time. The main contents and conclusions are as follows.

- (1) For the reinforcement learning process, an observation model of the balloon is built in accordance with the mission task of balloon station maintenance. Based on historical wind data, a wind field model with uncertainty is developed to present the transition probability of the RL framework. The framework is based on the Dueling Deep Double Q learning method. The ideal weight setting is determined by the control effects of numerous neural networks, and the matching neural network is derived as the optimal controller.
- (2) The appropriate definition of reward function and influence of weight distributions are discussed and compared. According to the control effects of multiple neural networks, the optimal weight setting is selected, and the corresponding neural network is obtained as the optimal controller.
- (3) The distribution characteristics of the quasi-zero wind layer in the specific region are summarized, and a variable wind field modeling method is used to train and test the robustness of the controller. According to the statistics of total reward, horizontal displacement, and effective working time, the result is that the performance of controller decreases with the increase of the wind uncertainty. When the bandwidth parameter κ is less than 6 , the performance degradation of the controller is obvious.
- (4) The effect of the initial point is examined to select a suitable altitude for the balloon deployment. Compared with the altitude with a high wind velocity, the performance of the controller is better at the altitude with a lower wind velocity. Furthermore, the influence of wind uncertainty is more pronounced at high wind velocity altitudes than at low wind velocity altitudes.

These results provide a valuable reference for balloon deployment. According to the wind field prognostic model and reanalysis data, the launch time can be selected based on the ability of the controller. As the balloon can fly for a long time, it can easily cross different regions. The solution of the position control under the temporal and spatial variable wind field may be considered in the next phase of the work.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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